

Title

Structured Nursing Intervention and Recovery Outcomes in Hip Arthroplasty Patients

Abstract

Ninety older adults (mean age 67 years) undergoing total hip replacement were randomized to receive either an enhanced recovery nursing protocol (ERAS, n = 45) or standard postoperative care (n = 45). The ERAS group received optimized pain management, early mobilization, catheter removal protocols, nutritional support, and other multimodal interventions. The ERAS group demonstrated earlier return of gastrointestinal function (mean 1.2 days vs 2.1 days, $p = 0.004$), earlier independent ambulation (2.4 vs 3.8 days, $p = 0.002$), and faster catheter removal (1.8 vs 2.9 days, $p = 0.008$). Pain scores were significantly lower in the ERAS group across all postoperative timepoints (mean reduction of 1.8 points on 10-point scale, $p < 0.001$). Hospital length of stay was reduced by approximately 1.5 days in the ERAS group (5.2 vs 6.7 days, $p = 0.009$). Functional hip scores at discharge were higher in the ERAS group (mean 58 vs 51 points, $p = 0.03$). Postoperative complication rates were lower in the ERAS group (18% vs 31%, $p = 0.08$). Patient satisfaction ratings were significantly higher in the ERAS group (mean 8.6/10 vs 7.2/10). These findings demonstrate that structured nursing protocols substantially improve recovery trajectory and reduce hospitalization burden in older hip replacement patients.